MILITARY SPECIFICATION

BAROMETER, ANEROID ML-434( )/PM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of precision barometer used for meteorological purposes and designated Barometer, Aneroid ML-434( )/PM.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-387 - Plastic Sheet, Laminated, Thermosetting, for Designation Plates.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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MIL-P-116	- Preservation, Methods of.
MIL-P-5425	- Plastic, Sheet, Acrylic, Heat Resistant.
MIL-P-11268	- Parts, Materials, and Processes used in Electronic Communications Equipment.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes for Ground Signal Equipment.
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-169	- Extreme-Temperature Cycle.
MIL-STD-170	- Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods for Aerospace and Ground Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ELECTRONIC COMMAND

SC-DL-106482	- Barometer, Aneroid, ML-434( )/PM.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

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3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.2d).

3.2 Inspection standards. Prior to submission of any samples for inspection, the contractor shall submit a complete description of the prime standards and the test standards he proposes to use in the inspection of barometers for the approval of the contracting officer.

3.2.1 Prime standard. The prime standard shall be a mercurial manometer that has been calibrated by a standard of accuracy which is traceable to the National Bureau of Standards. The calibration shall be throughout the entire pressure range of the barometer. Three certified copies of the most recent calibration shall be submitted to the Government for approval by the contracting officer.

3.2.2 Test standard. The test standard may be either a mercurial or aneroid manometer. The manometer, when compared with the prime standard on two successive runs, shall be able to repeat readings at the same points throughout the range within 0.2 millibars. Three certified copies of the most recent comparative runs shall be submitted for approval by the contracting officer. Certified copies of this data shall be made available to the Government.

3.2.3 Rechecking the test standard. The calibration of the inspection test standard against the prime standard shall be rechecked throughout the test range for which it will be used, before making any inspection, and at intervals not exceeding 1 month during the time it is used for inspection.

3.3 Construction. The equipment shall be constructed in accordance with the requirements of this specification and SC-DL-106482.

3.4 Parts, materials, and processes. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.5 Recycled, virgin and reclaimed materials. Without jeopardizing the intended use of equipment furnished to this specification, selection of materials shall be governed by the following:

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a. There is no exclusion to the use of recovered materials.
b. There is no requirement that equipment be manufactured from virgin materials.
c. Reclaimed materials are required to the maximum extent possible.

3.6 Finish. The equipment shall be finished in accordance with MIL-F-14072 and SC-DL-106482.

3.7 Marking. Marking shall conform to MIL-M-13231 and SC-DL-106482.

3.7.1 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.2 Serial numbers. Each Barometer, Aneroid ML-434( )/PM shall have a serial number.

3.8 Fungus. Equipment and assemblies shall show no evidence of viable fungus or corrosion when subjected to the test in accordance with 4.17. (Note: Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.)

3.9 Interchangeability. Like units, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.18).

3.10 Details of components.

3.10.1 Aneroid unit. The aneroid unit shall consist of an evacuated metal cell, or cells, sensitive to variations in atmospheric pressure. The cell shall be connected to an indicating pointer by a linkage mechanism in such a manner that the movement of a flexible surface or surfaces of the cell, for a given change in atmospheric pressure, will be magnified by the linkage mechanism and transmitted to the indicating pointer with a minimum of friction in the moving parts and a minimum of distortion other than that which may be intentionally introduced to obtain desirable calibration characteristics.

3.10.2 Pressure sensitive cell.

3.10.2.1 Structure (soldered joints). The structure of the cell shall be such that soldered joints are subjected to a minimum of binding and shearing stresses; or, if this condition is

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not attained, the area of the soldered joints shall be sufficiently large to insure that cold-flow of the solder under strain will not be sufficient to cause an appreciable deformation.

3.10.2.2 Shearing stresses. Shearing stresses in limited areas of the flexible surfaces of the cell shall be eliminated as nearly as possible. To accomplish this, the structure of the cell shall be such that the forces of any balancing spring that may be used will be symmetrically applied to the flexible surfaces of the cell and will be uniformly distributed over these surfaces.

3.10.2.3 Balancing spring. The internal balancing spring shall maintain the upper and lower surfaces of the cell sensibly parallel throughout the entire pressure range. Cells in which the pressure of the spring is unevenly distributed, causing the cells to lean or tip to one side, will not be acceptable. The cell shall be so constructed that the balancing spring cannot change its position on the surfaces on which it bears, as the cell expands or contracts. The balancing spring shall not come in contact with the flexible surfaces of the cell at any place throughout the pressure range of the instrument. The movement shall be smooth and uniform throughout the pressure limits of the cell.

3.10.3 Linkage mechanism.

3.10.3.1 Mounting. The pressure cell shall be mounted with one end of the cell attached to the mounting frame so that its position relative to the frame can be altered only by intentional adjustment. The other end of the pressure cell shall be attached to the linkage mechanism and shall be free to move in response to changes in the dimensions of the cell caused by variations in atmospheric pressure.

3.10.3.2 Balance. The linkage mechanism shall be so constructed that the instrument will comply with the test of 4.8. The linkage mechanism shall be accurately balanced or counterbalanced, or maintained under sufficient spring tension, to insure that unbalanced weights will not cause shifts in the position of the indicating pointer greater than the tolerance specified in 4.8.

3.10.3.3 Fit. The fit shall be such as to produce a mechanism with a minimum of friction, looseness and backlash. The indicating pointer shaft and all other shafts in the system shall have jeweled bearings throughout. The mechanism shall be constructed to preclude any disengagement of any of the elements thereof.

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3.10.3.4 Range. The range shall be such that a total pressure variation of not less than 300 millibars will be indicated. The pressure variation shall actuate the pointer through 690 ± 30 circular degrees.

3.10.4 Adjustment mechanisms. A means shall be provided to enable the indicating pointer to be set to read current atmospheric pressure as indicated by a standard mercurial barometer. Construction shall be such that the adjustment cannot be changed unintentionally or by severe vibration. One adjustment shall be provided for large scale adjustments, another shall be provided for accurately making very small changes in the indication of the instrument. At the contractor's option, these two adjustments may be incorporated in the same construction. The large scale adjustment shall be capable of being made without varying the calibration characteristics of the instrument. This requirement shall also apply where the two adjustments are incorporated in the same construction.

3.10.4.1 Adjustment facility. It shall be possible to adjust the mechanism with a screwdriver from outside the instrument. The slotted screwhead shall be at least large enough to accommodate a 0.22 inch (5.56mm) wide screwdriver blade. The width of the slot shall be at least 0.030 inch (0.76mm). Construction shall insure that access to the adjusting mechanism or adjustment thereof cannot damage other parts of the instrument. Adjustment shall be made through either the side or the front of the instrument case, at the contractor's option. No part of the adjusting means shall project outside the instrument case. Means which require adjustments to be made through the back of the instrument case, or require the position of the instrument to be changed during adjustment, will not be acceptable.

3.10.5 Indicating shaft and pointers.

3.10.5.1 Position. The indicating pointer shaft shall be perpendicular to the plane of the indicating dial. The shaft shall project above the surface of this dial sufficiently to mount the pointer, with at least 0.079 inch (2.01mm) of clearance at any position of rotation.

3.10.5.2 Pointer. The indicating pointer shall be straight, light in weight, and accurately balanced with reference to its point of attachment on the pointer shaft. The pointer shall be made of light color metal that will contrast with the finish of the sighting edge. That part of the pointer which projects over the graduated scales of the indicating dial shall have a width in a cross-section parallel to the dial surface of 0.001 to 0.004

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inch (.025 to .102mm). The pointer shall not be permanently deformed in shape by severe jarring of the instrument. The construction shall insure that the fit of the pointer on the shaft will not be altered by repeated removal and replacement of the pointer. The method of attachment shall be such that the indicating pointer can be removed without the use of special tools. The sighting edge of that portion of the pointer that projects over the graduated portion of the dial shall be finished in a dark color. The sides of the portion of the pointer which are perpendicular to the dial shall not be painted. The end of the pointer shall extend at least as far as the outside diameter of the graduations.

3.10.5.3 Auxiliary pointer. An auxiliary pointer shall be provided to indicate the significant portion of the scale.

3.10.6 Mounting frame.

3.10.6.1 Mounting frame. A mounting frame shall be provided to which shall be attached the entire mechanism and the indicating dial. The indicating dial may be an integral part of the mounting frame provided the adjustment means furnished in accordance with the requirements of 3.10.4 do not require a rotatable dial.

3.10.6.2 Rigidity. The mounting frame structure shall be sufficiently rugged so that all parts of the aneroid unit will be maintained in alinement without changes in calibration greater than the errors permitted in section 4 after the instrument has been subjected to the vibration test specified in 4.13.

3.10.6.3 Movement limitation. If a suspension structure is employed, the movement of the mounting frame in this suspension shall be limited to 0.197 inch (5.0mm) by means of at least three stops spaced uniformly around the frame.

3.10.7 Indicating dial.

3.10.7.1 Face. The face of the dial shall have a white matte finish, contrasting well with the black graduation lines. The dial shall meet the tests specified in 4.15 and 4.16. The diameter shall be between 2.640 and 2.758 inches (67.1 and 70.1mm). Each dial shall be hand-calibrated for each instrument to have maximum accuracy with the barometer in a horizontal position.

3.10.7.2 Graduation. All lines denoting a particular division of the scale, such as one, five, or ten millibar divisions, respectively, shall be approximately the same length; however,

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the lines denoting five millibars shall be 50 percent longer than the lines denoting one millibar and lines denoting ten millibars shall be 50 percent longer than the 5 millibar lines. Line width of scale markings shall be 0.004 to 0.006 inch (.102 to .152mm). The letters, figures, and the spacing of scales relative to each other shall be satisfactory for ease of reading. All graduation lines shall be portions of radii extending from a common point on the axis of the needle shaft, and shall be of uniform width.

3.10.8 Instrument case.

3.10.8.1 Window. A window slightly larger in size than the indicating dial shall be provided in the case as a cover for the dial. The window shall be of such size that all figures and graduations on the dial will be visible when viewed from any point directly above the window. The material of the window shall be a plastic in accordance with finish A of MIL-P-5425. A construction shall be used which will permit replacement of the window without special tools. There shall be a clearance of not less than 0.125 inch (3.175mm) between the indicating pointer and the window. The thickness of the window shall be such that, in normal handling, the window cannot be deformed to a point where it can touch the pointer. The window shall be securely attached to the instrument case. Means of attachment employing friction fits will not be acceptable. It shall be possible to gain access to the aneroid mechanism by removing the window, or a member in which the window is mounted.

3.10.8.2 Vent. The instrument case shall have a vent, or valve, which, when in the open position, will allow sufficient interchange of air between the inside of the case and the atmosphere to permit immediate response by the barometer to any change in atmospheric pressure. When the vent is closed, the instrument case shall be airtight to meet the test of 4.11.

3.10.9 Stops. Stops shall be provided to limit the travel of the indicating pointer to the minimum and maximum readings on the scale.

3.10.10 Correction chart. The contractor shall supply a single chart, containing the information specified in the following subparagraph. The chart material shall consist of white paper of book grade laminated between two sheets of clear transparent plastic material, type GCP-H in accordance with L-P-387. The two sheets, each with a minimum thickness of 0.015 inch (0.381mm), shall be bonded together to seal against moisture.

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3.10.10.1 Calibration. A chart shall be furnished with each instrument from which it shall be possible to obtain temperature and scale calibration corrections to be applied at any pressure within the operating range. The data for this chart shall be obtained from each barometer individually; data giving the average performance for a group of instruments will be unacceptable.

3.11 Service conditions. The barometer shall meet the following service conditions:

3.11.1 Barometer (operating). The barometer shall meet the requirements of this specification under any probable combination of the following:

a. Temperature. Any ambient temperature in the range of -40°F to 150°F (-40°C to 66°C), including sun load.

b. Relative humidity. Relative humidity up to 95 percent.

c. Elevation. Up to 7,000 feet (2134m) above sea level.

3.11.2 Barometer (non-operating). The barometer shall be capable of meeting the requirements of 3.11.1 after subjection to any one or more of the following non-operating conditions:

a. Temperature. Continuous exposure in the range of -80°F to 160°F (-62°C to 71°C) for 72 hours at any orientation with vent open.
b. Relative humidity. Relative humidity up to 98 percent for 96 hours, including condensation caused by temperature changes,
c. Elevation. Elevation up to 30,000 feet (9144m) above sea level with the vent closed.
d. Vibration. The barometer shall satisfactorily pass the test of 4.13.
e. Fungus. Barometer must satisfactorily pass the test of 4.17.

3.12 Facsimiles. Facsimile copies of the correction chart and the indicating dial shall be submitted to the contracting officer for approval (see 6.2g).

3.13 Preconditioning. The barometer shall be capable of meeting the inspection of section 4 without subsequent processing, after subjection to the bounce preconditioning of 4.6.

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3.14 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2i).

3.15 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268:

- Rubber	- Tropicalization of materials.
- Springs.	- Adhesives.
- Cleaning of parts.	- Thermoplastics.
- Soldering.	- Plastic Materials & Parts.
- Securing of parts.	- Screws, other threads.
- Sealing compounds.	- Devices, and related parts.

b. In this specification:

Pressure sensitive cell (3.10.2).
Balancing spring (3.10.2.3).
Mounting frame (3.10.6).
Fit (3.10.3.3).
Indicating dial (3.10.7).
Instrument case (3.10.8).
Adjustment facility (3.10.4.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3) (does not include packaging).
b. Inspections covered by subsidiary documents (see 4.4).

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c. Quality conformance inspection.
(1) Quality conformance inspection of equipment before packaging (see 4.5).
(2) Quality conformance inspection for packaging (see 4.19).

4.3 First article inspection. The first article inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for group A, group B, and group C testing.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirements specified in the contract.

Item	Requirement paragraph
Finish	3.6
Marking	3.7

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. This inspection, including sampling, shall conform to table I and the normal inspection procedures of MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (% defective)	
			Major	Minor
Visual and mechanical	3.15	4.7	4	6.5

4.5.2 Group B inspection. This inspection, including sampling, shall conform to table II and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection level S-4. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples

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selected from units that have been subjected to and met the group A inspection.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (% defective)	
			Major	Minor
Position effect	3.10.5	4.8	1.0 for the group	1/
Linkage-backlash and friction	3.10.3	4.9		
Scale calibration	3.10.4	4.10		
Leakage	3.10.8.2	4.11		
Interchangeability	3.9	4.18	4.0	6.5

1/ All operating defects are classified as major.

4.5.3 Group C inspection. This inspection shall consist of the tests specified in table III and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Hysteresis	3.10.2, 3.10.3	4.12
Temperature	3.11.1a, 3.11.2a	4.15
Moisture resistance	3.11.1b, 3.11.2b	4.16
Fungus	3.11.2e	4.17
Altitude	3.11.1c, 3.11.2c	4.14
Vibration	3.11.2d	4.13

4.5.3.1 Sampling for inspection. Two sample units shall be selected at random from the first 50 items produced and one from each 50 produced thereafter or portion thereof.

4.5.3.2 Noncompliance. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the

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proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Bounce preconditioning. The unit shall be placed in its normal operating position on the table of the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a strip of material, 0.03125 inch (0.8mm) thick under one corner or edge of the unit to a distance of 3 inches (76.2mm) as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance testing.

4.7 Visual and mechanical inspection. The barometer shall be inspected for workmanship, mechanical fit, including screw fits, fit of needle on shaft, quality of materials, markings, and miscellaneous defects. Finishes shall be checked for satisfactory workmanship. Clearances, dimensions and mechanical adjustments shall be checked. The correction chart and indicating dial shall be inspected for conformance with approval facsimiles. Visual and mechanical defects shall be classified as major or minor in accordance with the definitions of MIL-STD-105.

4.8 Position effect. Readings shall be taken of the barometer when placed such that the dial is first in a vertical position and then in a horizontal position. The difference in indication between these two positions, when taken at the same elevation, shall not exceed 0.3 millibar on the pressure scale. With the dial in a vertical position, the instrument shall be rotated slowly through 360 degrees in any plane, at the same elevation. The difference in indication between any two positions shall not exceed 0.5 millibar.

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4.9 Friction, looseness and backlash. The following test for friction, looseness and backlash in the linkage mechanism shall be applied:

a. The barometer shall be held in the hand and carefully read. Without jarring or in any way rotating the dial, the barometer shall be carefully lowered a distance of approximately 6 feet (1.83m). There shall be a perceptible increase in the indication of the pointer. This test shall be repeated except that the instrument shall now be raised approximately 6 feet (1.83m). There shall be a perceptible decrease in pointer indication.
b. The instrument shall be subjected to a pressure change of 0.3 millibar. The indication of the pointer shall show a change in the direction of the pressure change without vibrating the instrument.
c. The movement of the indicating pointer during the test specified in 4.10 shall be carefully noted. When the pressure is changed uniformly, there shall be no evidence of uneven motion or hanging of the pointer at any point on the dial. Irregular motion of the indicating pointer at any point on the scale, with uniformly varying pressure shall be considered cause for rejection of the instrument.

4.10 Scale calibration.

4.10.1 Test procedure.

a. The instrument shall not be subjected to pressure variations in excess of 1 inch (25.4mm) of mercury for a period of 15 hours preceding each test.
b. In making temperature tests, the instrument shall be held for a period of at least 1 hour preceding the test within $\pm 5^{\circ}F$ ($2.8^{\circ}C$) of the temperature at which the test is to be made. During the test, the temperature shall be held within $\pm 2^{\circ}F$ ($1.1^{\circ}C$) of its value at the beginning of the test.
c. Leakage shall be eliminated or reduced to a minimum in test equipment. When leakage is present, it shall be in the same direction as the pressure change under observation at the time data is recorded.

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d. When subjecting the instrument to changes in pressure, the pressure shall first be reduced from current atmospheric pressure to the low pressure limit of the test, then increased to the upper pressure limit of the test, and finally reduced to atmospheric pressure. Unless otherwise required, the lower pressure limit of all tests shall be not greater than 750 millibars pressure and the upper limit not less than 1050 millibars pressure. Test data shall be taken at both increasing and decreasing pressures.

e. Unless otherwise required, test points shall be taken at not greater than 30 millibar intervals. The test points, for increasing and decreasing pressures, shall be maintained to within 1 millibar of desired value for any one test cycle.

f. The rate of pressure change between test points shall be not more than 50 millibars per minute. The pressure shall be held approximately constant for at least 5 minutes prior to obtaining test data. Scale calibration tests shall be made with the barometer dial in a horizontal position.

g. Where reference is made to the error permitted in a test, this error shall be defined as the difference between the indication of the barometer under test and the corrected indication of the standard manometer when data from both instruments are taken simultaneously. In noting the indication of the barometer under test the instrument may be lightly tapped or vibrated; then the position of the indicating needle shall be estimated as closely as possible. The indications of the test manometer shall be carefully determined to the nearest 0.05 millibar.

4.10.2 Calibration criteria. The barometer shall be tested at the temperature shown in table IV. Each test shall cover the entire range of the barometer in accordance with the preceding test procedure. Test number 2 is a repeatability test. The comparative readings between the instrument and the standard manometer shall not exceed the allowable errors indicated in table IV. At the same pressure point, the difference in error between data taken at increasing and decreasing pressures shall not exceed 0.8 millibar at any point in the test range.

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TABLE IV. Calibration criteria.

Test number	Test temperature		Allowable error (millibars)	
	°F	°C	Basic data	With chart
1	75	23.9	±1.0	±0.5
2	75	23.9	±1.0	±0.5
3	140	60	±2.0	±0.5
4	-40	-40	±2.0	±0.5

4.11 Leakage. The vent of the case shall be closed at ambient atmospheric pressure and the case shall be subjected to a reduced pressure of 115 ±10 millibars. The reduction of pressure within the case, due to leakage, shall not exceed 10 millibars in 3 hours. The temperature shall remain constant to within ±2°F (1.1°C) throughout the test.

4.12 Hysteresis. The barometer shall not be subjected to pressure variations in excess of 1 inch (25.4mm) of mercury for a period of 15 hours preceding this test. The instrument shall be compared with a standard manometer at room pressure, and shall then be subjected to a pressure of approximately 750 millibars for a period of at least 15 hours. During this interval the pressure shall not vary by more than 50 millibars. The temperature to which the instrument is exposed shall be room temperature. At the end of this interval, the pressure shall be increased to room pressure in not more than 5 minutes time, and the instrument read within one minute. The difference between the error before and the error after this test shall not exceed 1.2 millibars. After a period of 8 hours this difference in error shall not exceed 0.3 millibar.

4.13 Vibration. The barometer shall be clamped, in the required positions, on a vibrating table that can be controlled to within 10 percent of the specified frequencies and amplitudes. The vibration table shall be designed to provide sinusoidal vibration at all specified frequencies and amplitudes.

4.13.1 Directions of vibration. The barometer shall be vibrated successively over the specified ranges of applied frequencies, in 3 mutually perpendicular directions. It shall be vibrated for at least 30 minutes in each of these directions (total of at least 90 minutes). The range of the applied frequencies shall be from 10 to 33 cycles per second.

4.13.2 Rate of change of vibration frequency. The frequency of applied vibration shall be varied uniformly at a rate of approximately 1 cycle per second per minute.

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4.13.3 Amplitude of applied vibration. The amplitude (one half of the total excursion) of the vibration table shall measure 0.020 inch (0.508mm).

4.13.4 Criterion of compliance. Upon completion of the vibration test, the tests specified in 4.10 shall be repeated and comparison of performance characteristics made with initial performance test results. The repeat tests shall be within the specified tolerances.

4.14 Elevation test. The equipment, placed in an altitude chamber in a normal operating position, shall be tested as follows:

a. Compare barometer reading with manometer, at ambient pressure, to establish a standard for the accuracy of instrument under test.
b. Reduce the barometric pressure in the test chamber to 23.1 ±0.1 inch (586.7 ±2.5mm) of mercury (simulated 7,000 feet (2134m) above sea level) and, after temperature and pressure stabilization, check barometer for accuracy of pressure reading.
c. With the barometer prepared for transport, place in a pressure chamber and reduce the barometric pressure to 8.9 ±0.1 inches (226 ±2.5mm) of mercury to simulate 30,000 feet (9144m) above sea level. Maintain this condition for 1 hour.
d. Return the chamber to ambient pressure. The barometer shall be checked for accuracy at no less than five cardinal points (both increasing and decreasing pressure) throughout the entire operating range.

4.15 Temperature test. The equipment shall be subjected to the temperature cycle shown in MIL-STD-169 and shall be in accordance with the following:

a. Take performance measurements to establish a standard of performance for the equipment before placing it in the temperature chamber.
b. During the temperature cycle, make performance checks at 150 ±5°F (66 ±2.8°C) and at -40 ±5°F (-40 ±2.8°C).
c. Subject the barometer to the extreme temperature and time duration in accordance with the temperature cycle test.
d. After the cycle has been completed, repeat the performance measurements taken in step "a"; repeatability of performance should be attained.

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4.16 Humidity test. The equipment shall be tested in accordance with MIL-STD-170 and the following:

4.16.1 Apparatus: humidity chamber and associated equipment. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be vented to the atmosphere to prevent the buildup of total pressure. Relative humidity shall be determined from the dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring agency. When readout charts are used, they shall be capable of being read with resolution within 1°F (0.56°C). When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 900 feet (274m) per minute. Provisions shall be made for controlling the flow of air throughout the internal test chamber area where the velocity of air shall not exceed 150 feet (45.7m) per minute. Steam, or distilled, demineralized, or deionized water having a pH value between 6.0 and 7.2 at 73°F (23°C) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the equipment by the test facility.

4.16.2 Standard ambient conditions. Standard ambient conditions are:

Temperature: 73°F ±5°F (23°C ±2.8°C)

Relative humidity: 50 percent ±5 percent

Atmospheric pressure: $28.5\ ^{+2.0}_{-3.0}$ inches of mercury ($724\text{mm}\ ^{+51\text{mm}}_{-76\text{mm}}$)

4.16.3 Installation of equipment in test facility. The equipment shall be installed in the test facility in a manner that will simulate service usage. Plugs, covers and inspection plates, not used in operation but used in servicing, shall remain in place. The test chamber shall be at standard ambient condition when the equipment is installed. Check the performance of the barometer by taking readings of atmospheric pressure as specified in 4.16.2. This reading shall be the operating standard for the barometer.

4.16.4 Test procedure.

Step 1. Place the equipment in the test chamber in accordance with 4.16.3.

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Step 2. Dry the test item at 130°F ±5°F (54°C ±2.8°C) for 24 hours.

Step 3. Condition the equipment at 73°F ±5°F (23°C ±2.8°C) and 50 ±5 percent relative humidity for 24 hours.

Step 4. Take initial measurements as specified in 4.16.5. Equipment shall meet full specification performance.

(Note: After step 4 the equipment may be realigned or readjusted as necessary to meet the specification requirements. No further realignment or readjustment shall be permitted throughout the test period. No repair or replacement of parts shall be permitted. Equipment shall not be operated except when specified test measurements are being performed and shall not be removed from the chamber for measurements.)

Step 5. Raise the internal chamber temperature to 86°F ±5°F (30°C ±2.8°C).

Step 6. Subject the equipment to 2 continuous 48-hour cycles in accordance with MIL-STD-170. Take measurements as specified in 4.16.5 and inspect the equipment in accordance with 4.16.6 at the period specified in the standard.

Step 7. After completion of step 6 cycling, condition the equipment for 24 hours at 73°F ±5°F (23°C ±2.8°C) and 50 ±5 percent humidity.

Step 8. Operate the equipment and compare performance with the data obtained in step 4.

Step 9. Inspect the equipment in accordance with 4.16.6 within an hour.

4.16.5 Performance. Check the performance under the same condition as specified in 4.16.2 and compare with the established standard. Repeatability should be achieved. Prior to performance test, accumulated moisture may be removed by inverting the equipment with gentle shaking. Wiping is not permitted.

4.16.6 Visual inspection and failure criteria. When specified, the equipment shall be visually inspected and a record made of any damage or deterioration resulting from the test. When

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practicable, visual inspection shall be performed within the chamber and at test conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external part which could in any manner prevent the equipment from meeting operational or maintenance requirements shall provide reason to consider the equipment as having failed the test.

4.16.7 Failure. If any equipment fails to meet the performance specified in 4.16.5 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet 4.16.5 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after the final specified conditioning and adjustment, it does not pass the test.

4.17 Fungus. The barometer shall be subjected to 28 days of fungus test specified in method 508 of MIL-STD-810. For the gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed with spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in the open condition.

4.18 Inspection for interchangeability. The external and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts and location of hinges and fasteners on separable parts or assemblies which must mate, shall be gaged or measured to determine conformance to physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

4.19 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 3P-1-1-ØØ-DC-NR-X-ED-Ø-00-A

5.1.2 Level B. 2M-1-1-ØØ-DC-NR-X-ED-Ø-00-B

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

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6. NOTES

6.1 Intended use. The Barometer, Aneroid ML-434( )/PM is a meteorological instrument used for accurately measuring pressure in the range from 750 to 1050 millibars. The barometer is contained in a metal case and indicates pressure by means of a pointer and circular scale which is graduated in intervals of 1 millibar.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.
b. Level A or B preservation and packing (see section 5).
c. When rough handling test is not required (see 4.19).
d. Number of first articles, if required, to be submitted for approval (see 3.1).
e. Place of final inspection.
f. Marking and shipping of samples.
g. Number of facsimiles required.
h. Contractor should apply for nomenclature in accordance with applicable clause in the contract.
i. If technical literature, tools, or spare parts are required.

6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: ML-434/PM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Verification inspection. Verification by the Government will be limited to the compliance with the provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-11839D Barometer, Aneroid ML-434( )/PM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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